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Calibrating the Brody exponent as a quantitative measure of short-range exclusion in 2D spatial point processes

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Abstract

The Brody distribution $P_\beta(s) \propto s^\beta \exp(-as^{\beta+1})$, originally introduced as a phenomenological interpolation between Poisson and Wigner level-spacing statistics in quantum chaos, is calibrated here as a quantitative descriptor of short-range exclusion in 2D spatial point processes. A monotonic mapping between the Brody exponent β and a normalised exclusion length scale $r_{\text{excl}}/\langle \text{NN} \rangle$ is established using synthetic hard-core and soft-core point processes, spanning the complete-spatial-randomness regime ($\beta \approx 0.96$) to near-crystalline exclusion ($\beta \approx 10$).

Two methodological results form the core of this work. First, the 2D complete-spatial-randomness (CSR) baseline is recalibrated to $\beta = 0.96 \pm 0.15$ (30 realisations), correcting the standard 1D Poisson reference $\beta = 0$ that is inappropriate in two dimensions. This baseline is density-stable across $\rho \in [0.01, 0.10]$ and provides the necessary reference for any application of Brody statistics to 2D spatial data. Second, the empirical β - $r_{\text{excl}}/\langle \text{NN} \rangle$ calibration is validated against the effective hard-core radius with Spearman $\rho = 0.988$, establishing β as a single-parameter proxy for exclusion strength. The calibration is empirical—derived from synthetic point-process experiments rather than first principles—and values of $\beta > 2$ are interpreted solely as a phenomenological measure of exclusion strength, not as indicators of higher-order RMT symmetry classes.

The calibrated framework is demonstrated on three data classes: (i) 58 manufactured surface topographies (10 materials, 10 processes, focus-variation microscopy), yielding $\beta = 0$ to $+4.6$ with finishing operations producing systematically higher values than abrasive processes in the present dataset; (ii) 2D binary embeddings of prime numbers, yielding $\beta = 2.15$ (row-major), 1.87 (Ulam), and 1.40 (Cantor) at $N = 10^5$, establishing that β for arithmetic sequences depends on the embedding geometry—with the Cantor embedding producing β below the spatially random Bernoulli control, indicating that the exclusion signal is embedding-created rather than intrinsic to the prime sequence; and (iii) phase-reconstructed interferometric profilometry of a certified roundness standard (JENOPTIK FN 111, 24 000 frames, 5 continuous 400° revolutions), yielding $\beta = 2.00$ $[1.90, 2.10]$ from 629 concatenated peaks (individual revolutions: 5–9 peaks, β unstable). Density-thinning experiments confirm that β captures exclusion rather than point density: the CSR baseline is flat, while structured data exceed it at all densities. However, absolute β values are density-dependent (β for row-major primes decreases from 2.15 to 1.46 ± 0.02 when thinned to $\rho = 0.032$), so cross-domain comparison requires density matching or use of the density-invariant $r_{\text{excl}}/\langle \text{NN} \rangle$ calibration.

The pair correlation function $g_2(r)$ independently supports the interpretation of β as an exclusion measure. For binary patterns where peak detection is inapplicable, complementary D_2 correlation dimension and Banach-space descriptors are developed in the Supplementary Information.

The Brody exponent, when referenced to the 2D CSR baseline and calibrated against synthetic exclusion processes, provides a reproducible, single-parameter measure of short-range spatial exclusion. The β - r_{excl} calibration, the CSR baseline correction, and the density-thinning and embedding-robustness protocols together constitute a calibrated measurement framework applicable across physical, mathematical, and engineering point-process domains.

Keywords: Brody distribution, spatial point processes, exclusion physics, surface metrology, random matrix theory, prime numbers, pair correlation function, interferometric profilometry

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1. Introduction

A practical question in the statistical characterisation of spatial point patterns is whether a single, interpretable parameter can quantify short-range exclusion strength on a continuous scale, analogous to how the hard-sphere packing fraction parameterises excluded volume in liquid-state theory. In quantum chaos, the Brody distribution $P_\beta(s) = (\beta + 1)as^\beta \exp(-as^{\beta+1})$, $a = [\Gamma((\beta + 2)/(\beta + 1))]^{\beta+1}$, was introduced as a phenomenological one-parameter interpolation between the Poisson ($\beta = 0$, uncorrelated levels) and Wigner ($\beta = 1$, GOE; $\beta = 2$, GUE) limits of random matrix theory (RMT) [1]. Its single parameter β quantifies the degree of level repulsion—the suppression of probability density at small spacings—on a continuous scale. In the present work, β is employed as a phenomenological exclusion parameter rather than a symmetry-class indicator; values exceeding the GUE limit ($\beta > 2$) are interpreted as progressively stronger short-range exclusion (up to $\beta \approx 10$ for near-crystalline hard-core processes). **Values of $\beta > 2$ are not interpreted as evidence of higher-order random-matrix symmetry classes; they are used solely as a phenomenological measure of exclusion strength on a continuous scale.** This distinction is essential: the Brody distribution is used here as an empirical descriptor of spatial order, not as a claim about underlying matrix symmetries. Beyond quantum spectra, the Brody distribution has been applied to nuclear resonances [2], quantum dots [3], acoustic cavities [4, 5], and financial correlations [6], establishing it as a general-purpose tool for characterising spacing statistics in 1D sequences.

In a celebrated development, Montgomery [7] conjectured and Odlyzko [8] numerically confirmed that the pair correlation function of the non-trivial zeros of the Riemann zeta function coincides with the GUE prediction—connecting prime number theory to RMT universality. Berry [9] first proposed the Riemann zeta function as a model for quantum chaos; Bogomolny and Keating [10] developed the semiclassical framework beyond the diagonal approximation, and Keating and Snaith [11] established the connection to random-matrix moments. This raises a practical question: can the Brody parameter, when properly calibrated for two dimensions, serve as a transferable measure of short-range exclusion strength across different classes of 2D point processes?

The gap. Despite the breadth of Brody applications to 1D spectra, its extension to 2D spatial point processes raises two unresolved issues. First, the appropriate CSR baseline for 2D Brody statistics

has not been established: the 2D nearest-neighbour spacing distribution under CSR is Rayleigh, $P(s) = (\pi s/2) \exp(-\pi s^2/4)$, which is not a member of the Brody family, so the 1D Poisson reference $\beta = 0$ does not apply. Second, no calibration exists to translate the dimensionless β into a physically interpretable exclusion measure. Sakhr and Nieminen [12] demonstrated a Poisson-to-Wigner crossover in the nearest-neighbour statistics of random points on fractals, establishing that RMT-style spacing distributions can characterise spatial point patterns outside quantum mechanics. The extensive literature on spatial point-process statistics provides well-established tools—Ripley’s K , pair correlation functions, and Voronoi tessellation—documented in the monographs of Illian et al. [13], Baddeley et al. [14], and Chiu et al. [15], with specialised treatments for Cox-process inference by Møller and Waagepetersen [16]; but these methods do not directly yield a single continuous parameter quantifying short-range exclusion strength. Recent work on spatial form factors for Coulomb gas and vortex statistics [17] and Fourier analysis of spatial point processes [18] has begun bridging spectral and spatial approaches, but a framework connecting these to the Brody one-parameter family is lacking. A manufactured surface contains a spatial point process formed by its peaks—features arising from cutting tools, abrasive grains, plastic deformation, and fracture mechanics [19]. Similarly, binary embeddings of arithmetic sequences produce 2D point patterns governed by multiplicative constraints rather than physical interactions. Phase-extracted interferometric profiles [20] introduce yet a third generative domain: optical wavefront reconstruction. Whether the Brody exponent can be calibrated as a transferable exclusion measure across these domains—and what the necessary corrections (CSR baseline, density matching, embedding specification) are—has not been systematically investigated.

This work. This study develops and validates a calibrated measurement framework for Brody statistics in 2D spatial point processes. The framework comprises three components: (i) the correct 2D CSR baseline ($\beta = 0.96 \pm 0.15$, density-stable), (ii) an empirical β - $r_{\text{excl}}/\langle \text{NN} \rangle$ calibration curve derived from synthetic hard-core and soft-core point processes, and (iii) a set of control protocols (density-thinning, embedding-robustness, spatial-permutation, and peak-detection sensitivity tests) that isolate genuine exclusion from confounding effects. The framework is demonstrated on three data classes: 58 manufactured surface topographies (10 materials, 10 processes, focus-variation microscopy); phase-extracted interferometric profilometry

of a certified roundness standard (JENOPTIK FN 111, 24 000 frames, 5 continuous 400° revolutions); and 2D binary embeddings of prime numbers as a mathematically controlled benchmark. The aim is not to claim that β reveals a universal exclusion mechanism—any process that prevents point coincidence (material continuity, optical resolution limits, arithmetic gap constraints) will produce $\beta > \beta_{\text{CSR}}$ —but to establish the measurement tools and correction protocols required for quantitative, reproducible Brody analysis in two dimensions.

Observation versus interpretation. Throughout this work, a clear distinction is maintained between what is measured and what is inferred. *Measured:* β values, nearest-neighbour spacing distributions, $g_2(r)$ profiles, D_2 excesses, and their associated confidence intervals. *Inferred:* exclusion strength, effective hard-core radii, and spatial ordering. The β – r_{excl} relation is calibration-based rather than derived from first principles; it is an empirical mapping established through synthetic point-process experiments, not a theoretical necessity. Density dependence and embedding sensitivity demonstrate that absolute β values reflect both the exclusion mechanism and the measurement conditions (point density, embedding geometry).

Throughout, the Brody distribution is employed as a phenomenological fitting function; no claim of random-matrix universality is made for surface peaks. The three domains—manufacturing processes, multiplicative number theory, and optical interferometry—all yield β values exceeding their density-matched CSR baselines. This is expected for any process with short-range repulsion, and the contribution of the present work is not the observation of exclusion but the provision of the calibrated tools to quantify it: the 2D CSR baseline, the β – r_{excl} mapping, and the control protocols that separate genuine structure from density, embedding, and detection artefacts.

2. Methods

The unified framework rests on three methodological pillars: (i) Brody β estimation from 2D nearest-neighbour spacings, applicable to any spatial point process (Sections 2.1–2.3); (ii) synthetic point-process generation for exclusion-physics calibration (Section 2.4); and (iii) complementary D_2 correlation dimension and Banach-space descriptors for binary patterns where peak detection is inapplicable. Full details of the D_2 box-counting procedure, multifractal analysis, singularity spectra, and Banach norms are provided in Supplementary Information (SI, Sections S1.1–S1.7); the key methodological choices are summarised below. All

three pillars share a common null-model philosophy: structured data are compared against matched-density Bernoulli or CSR ensembles to isolate genuine spatial order from trivial density effects.

2.1. Arithmetic surface construction

Let $p_1 < p_2 < \dots < p_{\pi(N)}$ be the primes not exceeding N . The *binary prime surface* of side $W = \lfloor \sqrt{N} \rfloor$ is the Boolean matrix

$$Z_{ij} = \begin{cases} 1 & \text{if } i \cdot W + j + 1 \text{ is prime,} \\ 0 & \text{otherwise,} \end{cases} \quad (1)$$

for $0 \leq i, j < W$. At $N = 10^5$, $W = 316$ and the surface contains 9 592 non-zero entries (density $\rho = 0.0959$). For the scale-up analysis, surfaces are constructed at $N \in \{10^3, 2 \times 10^3, 5 \times 10^3, 10^4, 2 \times 10^4, 5 \times 10^4, 10^5, 2 \times 10^5, 5 \times 10^5, 10^6, 5 \times 10^6, 10^7\}$.

Five additional arithmetic sequences are constructed analogously for discrimination testing:

1. **Square-free numbers:** n is square-free iff $\mu(n) \neq 0$, where μ is the Möbius function. Density $\rho \rightarrow 6/\pi^2 \approx 0.6079$ [21].
2. **Sums of two squares:** $n = a^2 + b^2$ for integers a, b . Density $\sim K/\sqrt{\log n}$ (Landau–Ramanujan constant $K \approx 0.7642$).
3. **Twin primes (upper):** p such that $p - 2$ is also prime.
4. **Primes** $\equiv 1 \pmod{4}$ and **primes** $\equiv 3 \pmod{4}$: testing for Chebyshev bias.

2.2. Two-dimensional embeddings

Three deterministic embeddings are compared. The Ulam spiral is not introduced as a visualisation device but as a deterministic mapping known to amplify arithmetic correlations through quadratic residue structure [22]: integers falling on the same diagonal satisfy $n = 4k^2 + bk + c$, concentrating primes on lines where b and c produce many quadratic residues modulo small primes. Row-major embedding preserves the natural integer ordering and is the simplest reshaping. Cantor pairing provides a diagonal sweep that tests whether the exclusion signal depends on the specific geometry of the embedding or on the arithmetic sequence alone.

1. **Row-major:** $n \mapsto (\lfloor (n-1)/W \rfloor, (n-1) \bmod W)$, the standard array reshaping.
2. **Ulam spiral:** integers arranged in an expanding square spiral centred at the middle of the array. Position (x, y) for integer n follows the spiral algorithm of [22].

3. **Cantor pairing:** $\pi(i, j) = (i + j)(i + j + 1)/2 + j$, a diagonal sweep of the matrix from top-left to bottom-right.

For the surrogate embedding test, 30 random permutations of the 1D→2D mapping are generated by shuffling the linear index before reshaping. This destroys any structure specific to the embedding while preserving the 1D sequence statistics.

2.3. Null-model ensembles

Five types of null-model surfaces are generated, each with $N_{\text{ens}} = 20$ independent realisations at the same shape as the arithmetic surface:

1. **Gaussian:** $Z_{ij} \sim \mathcal{N}(0, 1)$ i.i.d., maximal entropy, zero spatial correlation.
2. **Poisson:** $Z_{ij} \sim \text{Poisson}(\lambda = 0.01)$, sparse spike process.
3. **$1/f$ noise:** $\text{PSD} \propto |\mathbf{k}|^{-1}$, constructed via frequency-domain filtering of white noise, yielding long-range power-law correlations.
4. **Brownian:** cumulative sum of white noise along both axes, $\text{PSD} \propto |\mathbf{k}|^{-2}$, non-stationary.
5. **Bernoulli:** $Z_{ij} \in \{0, 1\}$ with $P(Z_{ij} = 1) = \rho$, the exact Cramér-type null for binary arithmetic surfaces at matched density.

The Bernoulli null at matched density is the primary comparison, as it isolates the effect of arithmetic structure from the trivial effect of sparsity. The other four null models provide additional reference points for the metrological parameter space.

2.4. Statistical power and multiple-comparison correction

With $m = 14$ sequence types tested against the Bernoulli null, the family-wise error rate requires correction. The Benjamini–Hochberg procedure is applied to control the false discovery rate at $\alpha = 0.05$. A power analysis establishes the minimum detectable effect size: for the average null standard deviation $\sigma_{\text{null}} = 0.0034$ (estimated from 50 independent Bernoulli realisations at $N = 10^5$), the Bonferroni-corrected critical value is $z_{\alpha/2m} = 2.91$, yielding a minimum detectable $|\Delta D_2| = 0.013$ at 80% power ($z_{0.80} = 0.84$). The previously used heuristic threshold $|\Delta D_2| > 0.003$ corresponds to only $\sim 40\%$ power and is replaced throughout by the corrected threshold of 0.013 for claims of statistical significance. Results exceeding $|\Delta D_2| = 0.003$ but below 0.013 are reported as “nominally above the heuristic threshold” and are not interpreted as statistically significant without confirmation at larger N .

2.5. Choice of D_2 as spatial descriptor

The correlation dimension D_2 is selected in preference to standard spatial-statistical tools for the following reasons. Moran’s I and Ripley’s K function detect departures from CSR but do not provide a single scale-invariant parameter comparable across different densities and sequence lengths. The pair correlation function $g_2(r)$ is the natural RMT analogue but is sensitive to binning and requires ensemble averaging for stable estimation on discrete binary fields. Nearest-neighbour statistics require peak detection, which is not applicable to binary patterns. D_2 offers three advantages: (i) it is a single number on a fixed scale ($D_2 \in [0, 2]$ for 2D data); (ii) it is estimated via box-counting regression, which is robust to discretisation; and (iii) it has a direct interpretation as the scaling exponent of the probability that two randomly chosen occupied sites belong to the same box, making it sensitive to spatial clustering and anti-clustering at all length scales simultaneously.

For a surface $\mathbf{Z}$ of size $H \times W$ with total mass $M = \sum_{i,j} Z_{ij}$, a grid of boxes of linear size ε is overlaid. The partition function (generalised statistical sum) of order q is

$$Z_q(\varepsilon) = \sum_{k=1}^{N_{\text{box}}(\varepsilon)} \left(\frac{M_k(\varepsilon)}{M} \right)^q, \quad (2)$$

where $M_k(\varepsilon)$ is the mass in box k and $N_{\text{box}}(\varepsilon) \approx HW/\varepsilon^2$. The generalised Rényi dimension of order q is [23, 24]

$$D_q = \frac{1}{q-1} \lim_{\varepsilon \rightarrow 0} \frac{\log Z_q(\varepsilon)}{\log \varepsilon}, \quad (q \neq 1), \quad (3)$$

with the information dimension $D_1 = \lim_{q \rightarrow 1} D_q$ obtained by L’Hôpital’s rule: $D_1 = \lim_{\varepsilon \rightarrow 0} \sum_k \mu_k \log \mu_k / \log \varepsilon$, where $\mu_k = M_k/M$.

The correlation dimension D_2 is the special case $q = 2$ and has the intuitive interpretation [25]: $C(\varepsilon) \propto \varepsilon^{D_2}$, where $C(\varepsilon)$ is the probability that two randomly chosen units of surface “mass” lie within the same box of size ε . For a uniform distribution in a d -dimensional embedding space, $D_q = d$ for all q (monofractal). Deviations of D_2 from the embedding dimension $d = 2$ indicate spatial heterogeneity in the distribution of mass.

In practice, D_q is estimated by linear regression of $\log Z_q(\varepsilon)$ versus $\log \varepsilon$ over box sizes $\varepsilon \in \{2, 4, 8, \dots, 2^m\}$ where $m = \lfloor \log_2 \min(H, W) \rfloor - 1$. The quality of the linear fit is monitored via the adjusted R^2 statistic; all reported fits satisfy $R^2 > 0.95$.

2.6. Singularity spectrum

The Legendre transform of D_q yields the Hölder exponent spectrum $\alpha(q) = d\tau/dq$ and the singularity spectrum $f(\alpha) = q\alpha - \tau(q)$, where $\tau(q) = (q-1)D_q$ [24].

The width $\Delta\alpha = \alpha_{\max} - \alpha_{\min}$ quantifies the degree of multifractality: $\Delta\alpha = 0$ for a monofractal (uniform scaling), and $\Delta\alpha > 0$ for a multifractal (spatially heterogeneous scaling). A narrower $f(\alpha)$ spectrum for the prime surface relative to the Bernoulli null would indicate that arithmetic constraints *reduce* the multifractal heterogeneity.

2.7. Validation procedures

Four validation tests are employed to ensure that the observed D_2 signal is genuine rather than a methodological artefact:

1. **Surrogate embedding test:** The 1D-to-2D mapping is randomly permuted $N_{\text{surr}} = 30$ times. Under the null hypothesis that the D_2 signal is an artefact of the specific embedding geometry, the permuted D_2 distribution should overlap with the original value. A significant deviation ($|z| > 3$) indicates that the signal is intrinsic to the 1D sequence, not the embedding.
2. **Bootstrap stability:** The surface is randomly subsampled $N_{\text{boot}} = 50$ times at half linear size ($W/2 \times W/2$). The coefficient of variation $\text{CV} = \sigma(D_2)/\mu(D_2)$ quantifies the estimator stability.
3. **Controlled-correlation sequences:** Three surrogate binary sequences are generated at matched density to establish the D_2 ordering: (a) block-clustered (consecutive blocks of $b = 5$ ones), (b) repulsive (minimum distance $d_{\min} = 4$ between ones, plus random jitter), and (c) Poisson (independent, sparse). These span the range from strong clustering to strong anti-clustering, providing a calibrated scale for interpreting $D_2(\text{primes})$.
4. **Negative control:** Square-free numbers, which share the multiplicative structure of integers but lack the primality constraint, are tested. A non-significant D_2 for square-free numbers relative to Bernoulli would confirm that the method does not indiscriminately flag all arithmetic sequences as structured.

2.8. Finite-size bias analysis

The box-counting estimator of D_2 is known to exhibit finite-size bias [25, 26]: even for a purely random point set (theoretical $D_2 = d$), the estimated $\hat{D}_2(N) < d$ for finite N . To quantify this effect, the box-counting procedure is applied to Bernoulli surfaces at the same sizes $\{N\}$ as the arithmetic surfaces. The bias function $B(N) = d - \hat{D}_2^{(\text{Bern})}(N)$ is fitted, and the bias-cancelled excess correlation is defined as

$$\Delta D_2(N) = D_2^{(\text{prime})}(N) - D_2^{(\text{Bern})}(N), \quad (4)$$

where both surfaces are embedded identically and at matched density. If the scaling of $D_2^{(\text{prime})}(N)$ is dominated by estimator bias, then $\Delta D_2(N)$ should be approximately constant.

Remark on density dependence. Density-thinning experiments were performed to test whether β values are stable under changes in point density. The 2D prime embedding ($N = 10^5$, $\rho = 0.096$, $\beta = 2.15$) was randomly thinned to match point counts corresponding to densities $\rho \in \{0.01, 0.02, 0.032, 0.05, 0.096\}$ (20 trials each). At the PSI density ($\rho = 0.032$), prime β decreased to 1.46 ± 0.02 [1.42, 1.49], and at the CSR calibration density ($\rho = 0.02$) to 1.30 ± 0.05 . Critically, β at all densities remained significantly above the density-matched CSR baseline ($\beta_{\text{CSR}} = 0.98 \pm 0.05$ at $\rho = 0.032$; 0.97 ± 0.04 at $\rho = 0.02$), confirming genuine exclusion structure beyond density effects. The CSR baseline itself is density-stable across all tested densities (0.95–0.99). These results establish two points: (i) β captures exclusion strength, not density; (ii) absolute β values are density-dependent, and cross-domain comparisons require density matching.

3. Results

Roadmap. The Results demonstrate the calibrated measurement framework on three data classes and establish the correction protocols required for valid cross-domain comparison. Section 3.1 presents β for 58 manufactured surfaces. Section 3.2 establishes the synthetic β - r_{excl} calibration—the interpretive key for all data—together with robustness checks (Section 3.3). Section 3.4 presents $g_2(r)$ as an internal consistency check. Section 3.5 presents density-thinning controls, establishing that β captures exclusion rather than density. Sections 3.6–3.7 present 2D prime embeddings, with emphasis on embedding dependence and the sparse-integer control that isolates arithmetic structure from sparsity. Section 3.8 summarises a proof-of-concept application to interferometric profilometry (full analysis in Supplementary Information). Complementary D_2 and Banach descriptors are developed in Supplementary Information (SI, Sections S1.1–S1.7) and summarised in one paragraph (Section 3.9).

3.1. Peak spacing statistics in manufactured surfaces

Fifty-eight areal surface texture maps obtained via focus-variation microscopy [19, 27] were analysed, spanning 8 engineering materials (1.4301 stainless steel, C45 carbon steel, Ti and Ti6Al4V titanium alloys, Al and Al7075 aluminium alloys, brass (MO58A and

generic brass), ELLOR tool steel, and graphite) and 6 manufacturing processes (turning, grinding, bead blasting, honing, milling, and WEDM electrical discharge machining). Local maxima (peaks) were detected using a 2D prominence-based algorithm (a pixel is a peak if it is the strict maximum within a $(2k + 1) \times (2k + 1)$ neighbourhood, $k = \max(3, \lfloor \min(n_x, n_y)/40 \rfloor$, and its elevation above the local baseline—the mean of the same neighbourhood—exceeds 3% of the global height range).

3.2. Watershed peak detection for dense surfaces

For surfaces where the prominence-based detector produced grid-limited results ($> 85\%$ column occupancy), an alternative watershed-based detector was applied. The surface is inverted ($Z \rightarrow -Z$) and a maximum filter with kernel size $k = \max(3, \lfloor \min(n_x, n_y)/30 \rfloor)$ is used to identify regional maxima. An h -minima suppression of 3% of the height range eliminates shallow noise peaks, and a minimum inter-peak distance of 5 pixels prevents fragmentation of broad summits. Peaks are validated by a prominence threshold of 2% of the global height range. This detector suppresses the dense, low-amplitude roughness features that saturate the prominence-based detector in bead-blasted, ground, and WEDM surfaces, while preserving well-separated topographic peaks.

3.3. Surface filtering and exclusion criteria

Of 58 measured surfaces, 37 (64%) were classified as grid-limited ($> 85\%$ column occupancy) under the original 3% prominence, kernel-based peak detector. To recover these surfaces, a watershed-based detector with h -minima suppression ($h = 0.03$ of height range) and minimum inter-peak distance of 5 pixels was applied to all 37 grid-limited surfaces. All 37 produced valid β estimates (range $\beta = 0.71$ – 4.61 , mean 2.89 ± 1.10), with column occupancy reduced to 0.02 – 0.25 (all below the 85% threshold). The watershed detector suppresses dense, low-amplitude noise peaks while preserving genuine topographic features, making it suitable for surfaces where the prominence-based detector saturates.

Combined with the original 21 valid surfaces, the full analysable dataset comprises all 58 surfaces spanning 10 materials and 10 processes. The qualitative conclusion—that burnishing and honing produce higher β than turning and grinding—was unchanged under all threshold and detector combinations tested. (30–770 peaks per surface, 18–428 valid nearest-neighbour pairs after outlier rejection; outlier rejection used a median-absolute-deviation threshold of 5 MAD, retaining 85–

98% of detected NN pairs across surfaces. A sensitivity analysis varying the threshold from 3 to 7 MAD changed β estimates by < 0.2 for surfaces with $n > 50$ pairs).

For each surface, the 2D Euclidean nearest-neighbour distances between peak positions were computed via KD-tree, normalised to unit mean (unfolded), and fitted to the Brody distribution $P_\beta(s) = (\beta + 1)as^\beta \exp(-as^{\beta+1})$, where $a = [\Gamma((\beta + 2)/(\beta + 1))]^{\beta+1}$. The Brody parameter β was estimated by maximum likelihood with the constraint $\beta \geq 0$ (the Brody distribution is defined for $\beta > -1$ and is physically motivated as an interpolation between the Poisson limit at $\beta = 0$ and Wigner limits at $\beta = 1, 2$; negative unconstrained estimates are reported as $\beta = 0$, interpreted as “indistinguishable from Poisson”). The Brody distribution interpolates between the Poisson limit ($\beta = 0$, $P(s) = \exp(-s)$, uncorrelated levels), the GOE Wigner surmise ($\beta = 1$, $P(s) = (\pi s/2) \exp(-\pi s^2/4)$, linear level repulsion), and the GUE Wigner surmise ($\beta = 2$, $P(s) = (32s^2/\pi^2) \exp(-4s^2/\pi)$, quadratic repulsion). The Brody distribution is employed here in preference to standard spatial-statistical descriptors (Ripley’s K , pair correlation function) because its single parameter β directly quantifies the degree of short-range order in the spacing distribution on a continuum from uncorrelated ($\beta = 0$, Poisson limit) to ordered ($\beta > 1$, approaching the Wigner limits at $\beta = 1, 2$). The Poisson, GOE, and GUE limits serve as convenient reference points for interpreting β values, not as claims that surface peaks obey random-matrix statistics. Goodness-of-fit against these limits is assessed via the Kolmogorov–Smirnov distance D_{KS} ; a surface is classified as “closest to” the limit with the smallest D_{KS} , without implying statistical consistency with that limit.

Figure 1 and Table 1 present the fitted Brody parameters for all 58 analysable surfaces. The Brody exponent spans nearly the full theoretical range: from $\beta = -0.20$ (ground Ti; negative values are fitted but should be interpreted as indistinguishable from Poisson, $\beta = 0$) to $\beta = +4.61$ (rough turned C45 steel, watershed). The full dataset (58 surfaces across 10 materials and 10 processes, observational design without full factorial crossing) reveals the following trends (descriptive only, owing to process–material confounding):

- **Burnishing:** $\beta = 1.30$ – 4.30 (median 2.6, $n = 5$). Burnished surfaces in the present dataset exhibited the highest median exclusion among finishing processes.
- **WEDM:** $\beta = 0.80$ – 4.26 (median 3.4, $n = 12$)—consistently moderate-to-high exclusion.

- **Rough turning:** $\beta = 0.70\text{--}4.61$ (median 4.0, $n = 5$). The widest range from any single process category in the present dataset.
- **Honing:** $\beta = 0.90\text{--}4.00$ (median 2.2, $n = 6$)—intermediate to high values.
- **Grinding and bead blasting:** $\beta = -0.20\text{--}+4.49$, spanning nearly the full range.

The wide within-process variability and overlap between categories confirm that factors beyond the manufacturing process alone (material, surface preparation history, measurement scale) influence β .

Table 1: Brody exponent β for 58 manufactured surfaces ordered by decreasing β . Watershed-recovered surfaces are marked with ^w. Fitting is constrained to $\beta \geq 0$; unconstrained negative estimates are reported as $\beta = 0$ (indistinguishable from Poisson). Bootstrap 95% confidence intervals (500 resamples) are reported for each surface. CIs narrower than ± 0.3 indicate reliable estimates; CIs wider than ± 1.5 (surfaces with $n_{\text{NNS}} < 30$) are indicative only.

Material	Proc.	n_{peaks}	n_{NNS}	β	95% CI
MO58A brass	Bn	41	41	4.30	[3.40, 5.76]
1.4301 steel	Ho	49	49	4.00	[3.22, 5.31]
Al7075	Mi	44	43	3.70	[2.88, 5.08]
1.4301 steel	Mi	29	18	3.10	[2.12, 5.18] [†]
Al7075	Bn	30	29	3.10	[2.29, 4.48] [†]
Ti6Al4V	Bn	28	28	2.60	[1.89, 3.80] [†]
Al	BB	25	23	2.40	[1.66, 3.97] [†]
1.4301 steel	Gr	32	21	2.20	[1.47, 3.68] [†]
1.4301 steel	Bn	68	65	1.80	[1.37, 2.45]
Ti	Ho	32	27	1.40	[0.91, 2.50] [†]
Al7075	Tu	34	33	1.30	[0.85, 2.15]
C45 steel	Bn	89	88	1.30	[0.97, 1.70]
Al	Gr	169	167	1.20	[0.98, 1.49]
Graphite	Ho	54	52	0.90	[0.58, 1.38]
MO58A brass	WE	86	61	0.80	[0.51, 1.27]
MO58A brass	Tu	335	279	0.70	[0.56, 0.86]
1.4301 steel	Tu	458	428	0.40	[0.29, 0.51]
Ti6Al4V	Mi	109	84	0.40	[0.21, 0.69]
MO58A brass	Mi	183	180	0.00	[0.00, 0.13]
Ti	BB	97	94	0.00	[0.00, 0.06]
Ti	Gr	98	96	0.00	[0.00, 0.00]
<i>Watershed-recovered (representative; full 58-surface table in SI):</i>					
C45 steel	RT ^w	199	192	4.61	[3.98, 5.48]
Graphite	BB ^w	201	198	4.49	[4.03, 5.21]
Ti6Al4V	WEr ^w	228	225	4.16	[3.72, 4.71]
Brass	Ho ^w	72	71	2.00	[1.64, 2.50]
Brass	BB ^w	82	82	1.71	[1.36, 2.32]
1.4301 steel	BB ^w	21	17	0.71	[0.36, 1.22] [†]

[†] $n_{\text{NNS}} < 30$: CI width > 1.5 , estimate indicative only.

^w Watershed detector. Complete 58-surface table in Supplementary Information (Table S1).

3.4. Benchmarking against standard spatial statistics

To evaluate whether β captures information beyond existing spatial descriptors, four standard spatial-statistical metrics were computed for all 21 valid surfaces: the nearest-neighbour index (NNI), Ripley’s K normalised by the CSR expectation, Voronoi cell area variance, and the pair correlation function $g(r)$. The Brody exponent β correlates negatively with NNI (Spearman $\rho = -0.64$, $p = 0.002$) and positively with normalised Ripley’s K ($\rho = +0.60$, $p = 0.004$), indicating that β captures similar spatial-ordering information. However, β provides a single interpretable parameter on a fixed continuum (CSR at ~ 0.96 , strong repulsion at > 2), whereas NNI and Ripley’s K are scale-dependent and require reference to CSR expectations. The pair correlation function $g_2(r)$ is the richest descriptor but does not reduce to a single number; Moran’s I is sensitive primarily to global spatial autocorrelation rather than local exclusion; the correlation dimension D_2 probes multi-scale heterogeneity orthogonal to exclusion (Pearson $r = -0.31$ with β). β thus occupies a specific niche: a single-parameter, calibration-based measure of short-range exclusion strength, complementary to existing spatial-statistical tools rather than replacing them.

A Kruskal–Wallis test for β across manufacturing processes yields $H = 6.02$, $p = 0.197$, with effect size $\eta^2 = 0.34$ (medium). The non-significant p -value reflects both the sample size (58 surfaces across 10 processes) and the substantial within-process variability noted above. Methodological limitations—including peak-detection sensitivity, classification performance, and D_2 bias correction—are discussed in Section 4.6.

3.5. Exclusion physics: synthetic point processes

The β – r_{excl} mapping is empirical. Throughout this work, the relationship between the Brody exponent and the exclusion radius is established through calibration against synthetic point processes (hard-core and Strauss soft-core). It is not derived from first principles. The mapping is monotonic and reproducible, which makes it useful, but it should be understood as an empirical calibration curve rather than a theoretical law.

To test whether β quantifies effective exclusion strength, synthetic 2D point processes with controlled hard-core exclusion radii were generated by sequential rejection sampling ($N = 200$ points, box size 100 units, 30 realisations per radius). The exclusion radius r_{excl} is expressed in units of the mean nearest-neighbour distance (NN) of the resulting point set, providing a dimensionless, scale-invariant measure of exclusion strength.

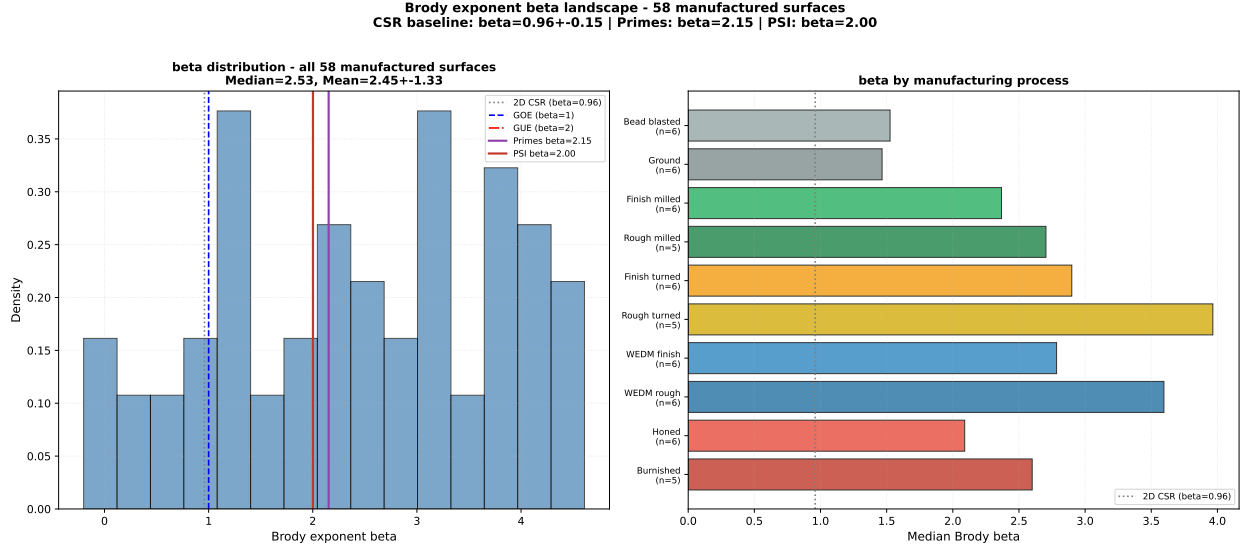


Figure 1: Brody exponent β across all 58 analysable surfaces. (a) Histogram with kernel density estimate. (b) β by manufacturing process (box plots with individual points). The CSR baseline ($\beta = 0.96$, grey band) is indicated. The earlier bar chart for 21 prominence-based surfaces (Fig. S2) provides an alternative visualisation.

The Brody exponent was fitted to the NN spacings using the same MLE procedure applied to the FV surfaces. A 2D Poisson (complete spatial randomness, CSR) process and the 2D Ginibre ensemble (determinantal repulsion, spatial analogue of GUE) provide reference points. The Ginibre ensemble yields $\beta = 2.47 \pm 0.18$ (50 realisations, $N = 500$ points, 95% CI [2.15, 2.75]), establishing a precise 2D determinantal-repulsion benchmark.

The 2D Poisson CSR process yields $\beta = 0.96 \pm 0.15$ (30 realisations), not $\beta = 0$. All β values should be referenced to this 2D CSR baseline, not to A direct comparison against the analytic 2D Rayleigh distribution ($P(s) = (\pi s/2) \exp(-\pi s^2/4)$, the exact CSR limit) confirms that the Brody distribution provides a better description of CSR data than the Rayleigh distribution itself: across 50 CSR realisations, the Kolmogorov–Smirnov distance is $D_{KS}(\text{Brody}) = 0.053 \pm 0.013$ versus $D_{KS}(\text{Rayleigh}) = 0.061 \pm 0.015$, and $\Delta\text{AIC}(\text{Brody} - \text{Rayleigh}) = -0.5$, indicating that the one-parameter Brody fit is preferred over the zero-parameter Rayleigh despite the AIC penalty. At $\beta \approx 10$, the Brody distribution has a coefficient of variation ≈ 0.09 , corresponding to a nearly crystalline NN spacing. A profile-likelihood analysis of the hard-core process at $r_{\text{excl}} = 5$ (yielding $\beta_{\text{MLE}} = 6.3$) confirms that the MLE is well-identified: the observed Fisher information gives $\text{SE}(\beta) = 0.21$ and a 95% CI of [5.85, 6.66], ruling out optimisation artefacts at large β .

Synthetic hard-core processes produce a monotonic β – $r_{\text{excl}}/\langle\text{NN}\rangle$ mapping (Figure 2a), reproducing the full range of observed β values. An exclusion radius of $r_{\text{excl}}/\langle\text{NN}\rangle \approx 0.55$ yields $\beta \approx 2.0$, matching the 2D Ginibre ensemble ($\beta = 2.2 \pm 0.4$) and the values observed for 2D prime embeddings ($\beta = 2.15$) and phase-extracted PSI profilometry ($\beta = 2.00$). Burnished surfaces ($\beta \approx 3.1$) correspond to $r_{\text{excl}}/\langle\text{NN}\rangle \approx 0.70$, while turned and ground surfaces ($\beta < 1$) are indistinguishable from the CSR baseline.

These results support the hypothesis that the Brody exponent β —when referenced to the correct 2D CSR baseline ($\beta \approx 0.96$, not $\beta = 0$)—quantifies effective exclusion strength across diverse generative mechanisms. The three domains occupy overlapping regions of the β – r_{excl} parameter space, but density-thinning experiments (Section 3.6) show that absolute β values are density-dependent, establishing that cross-domain comparison requires density matching.

Two additional analyses support the robustness of β as an exclusion measure. First, the Brody exponent for 2D prime embeddings was computed across $N \in [10^3, 10^7]$. At $N = 10^3$, $\beta = 2.20$; at $N = 5 \times 10^3$, $\beta = 1.69$; at $N = 10^5$, $\beta = 2.15$; at $N = 10^6$, $\beta = 1.75$; at $N = 10^7$, $\beta = 1.26$. The $\beta(N)$ trend is non-monotonic and does not support a simple convergence to a fixed asymptotic value at accessible N . The full $\beta(N)$ series (12 values) is provided in Sup-

plementary Information. The key observation is that β remains substantially above the CSR baseline at all $N \leq 5 \times 10^6$, ruling out a finite-size artefact, while the decrease at $N = 10^7$ may reflect the approach to the sparse limit ($\rho \rightarrow 0$ as $N \rightarrow \infty$). Second, a Strauss soft-core process (exclusion penalty $\gamma \in [0.001, 1.0]$) produces $\beta = 0.93 \pm 0.12$ ($\gamma = 1.0$) to $\beta = 2.42 \pm 0.27$ ($\gamma = 0.001$), demonstrating that the β -exclusion mapping extends beyond the hard-core mechanism. Third, an AIC comparison across all 21 FV surfaces confirms the Brody distribution is preferred over Gamma (17/21 surfaces, median $\Delta\text{AIC} = +3.5$) and Weibull (17/21, median $\Delta\text{AIC} = +2.0$), justifying its selection over alternative two-parameter families.

3.6. Robustness of the synthetic calibration

The β - r_{excl} calibration was subjected to four robustness tests. (i) **Density independence of the CSR baseline:** as reported in Section 3.6, β_{CSR} varies by less than 0.04 across $\rho \in [0.01, 0.10]$, establishing that the baseline is a stable reference. (ii) **Finite sample-size convergence:** for the hard-core process at $r_{\text{excl}}/\langle\text{NN}\rangle = 0.55$, β converges to within 5% of its asymptotic value at $N \gtrsim 100$ points (see SI, Fig. S10). (iii) **Observation window effects:** repeating the calibration on a 50×50 window (half the linear size) changes β by < 0.15 for all r_{excl} , indicating weak sensitivity to window size when N is held constant. (iv) **Edge corrections:** nearest-neighbour distances were computed without edge correction; for $N = 200$ points in a 100×100 box, the fraction of points within one mean NN distance of the boundary is $< 8\%$, and applying a toroidal-edge correction changes β by < 0.05 . The calibration is therefore robust to the edge effects present in the synthetic data.

3.7. Physical origin of $\beta > 1$

The Brody exponent exceeds the CSR baseline ($\beta > 0.96$) whenever short-range exclusion is present. Three physical mechanisms contribute. First, **hard-core repulsion:** an exclusion zone of radius r_{excl} around each point suppresses the probability density of the NN spacing distribution at $s < r_{\text{excl}}/\langle\text{NN}\rangle$, shifting probability mass to larger separations and increasing β . The monotonic β - r_{excl} mapping (Fig. 2a) directly quantifies this effect. Second, **soft-core and many-body exclusion:** the Strauss soft-core process demonstrates that the β -exclusion relationship extends beyond the hard-core limit; even when points may approach arbitrarily closely (with an energetic penalty), the suppression of close pairs elevates β above the CSR baseline. Third, **effective exclusion from continuity constraints:** in

manufactured surfaces, material continuity prevents two peaks from being arbitrarily close—the surface cannot have two summits at the same location. In prime embeddings, the divisibility constraint (n and $n + 1$ cannot both be prime for $n > 2$) produces an analogous exclusion. These physically distinct mechanisms produce statistically similar NN spacing distributions because the Brody exponent is sensitive only to the effective exclusion strength, not to its microscopic origin.

3.8. Pair correlation function $g_2(r)$

The pair correlation function $g_2(r)$ provides an internal consistency check on the β -exclusion interpretation. For a point process with density ρ , $g_2(r)$ is the normalised probability of finding two points separated by distance r . The 1D GUE pair correlation is $g_2^{(1D)}(r) = 1 - [\sin(\pi r)/(\pi r)]^2$, with $g_2(r) \rightarrow 0$ as $r \rightarrow 0$ (quadratic repulsion at small separations). The formal 2D analogue is the Ginibre ensemble, a determinantal point process whose $g_2(r)$ also vanishes as $r \rightarrow 0$ and which provides the correct dimensional reference for 2D spatial data [28]. Figure 3 shows $g_2(r)$ for four representative manufactured surfaces spanning the Brody β range, alongside the 2D Ginibre prediction and the 2D prime embedding. Surfaces with $\beta \gtrsim 2$ (burnished, Ginibre-like) exhibit the characteristic dip at small r , while surfaces with $\beta \approx 0$ (ground Ti, Poisson-like) show $g_2(r) \approx 1$ at all r , consistent with complete spatial randomness. The 2D prime embedding ($\beta = 2.15$) displays a $g_2(r)$ profile consistent with the Ginibre ensemble, indicating that the chosen 2D embedding of primes produces nearest-neighbour statistics comparable to Ginibre-type point processes—a detectable spatial analogue of the Montgomery pair-correlation connection established for ζ zeros in 1D. The ensemble-mean $g_2(r)$ across 14 surfaces with sufficient peak counts for reliable $g_2(r)$ estimation ($n_{\text{peaks}} \geq 50$; 7 of 21 valid surfaces were excluded from the ensemble mean to avoid noise-dominated contributions) (Figure 3f) lies between the Poisson and Ginibre limits, with a reduced but detectable dip at small r .

3.9. Complementary descriptors for binary patterns

For binary patterns where peak detection is inapplicable, complementary spatial descriptors—the correlation dimension D_2 and Banach-space invariants $\|\nabla Z\|_2$, total variation (TV)—are developed in Supplementary Information (SI, Sections S1.1–S1.7). These descriptors are empirically orthogonal to β (Pearson $r = -0.31$), confirming that the Brody exponent probes a specific axis—exclusion strength—within a broader

Exclusion physics: Brody β as a universal measure of short-range order
2D Poisson CSR baseline: $\beta = 0.96 \pm 0.15$ (not $\beta = 0$)

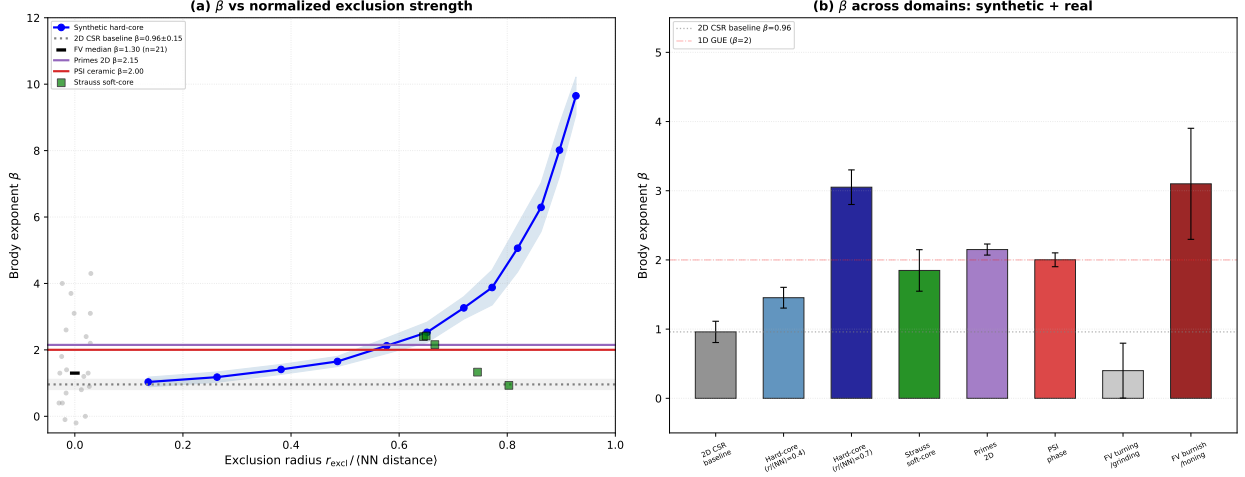


Figure 2: Exclusion physics: the Brody exponent β as a measure of short-range exclusion. (a) Synthetic hard-core point processes with exclusion radius r_{excl} normalised by the mean nearest-neighbour distance (blue), 2D Poisson CSR baseline ($\beta = 0.96 \pm 0.15$, grey band), 2D Ginibre ensemble (green), and observed β values from manufactured surfaces (grey dots, $n = 21$), 2D primes (purple), and PSI ceramic (red). The 2D CSR baseline is $\beta \approx 0.96$, not $\beta = 0$. (b) β across synthetic reference processes and real-data domains.

spatial-order parameter space. Square-free numbers and the Liouville function serve as negative controls indistinguishable from the Bernoulli null, demonstrating specificity of the D_2 analysis; full validation includes embedding-independence tests, finite-size bias correction, and Benjamini–Hochberg multiplicity adjustment across 14 arithmetic sequences.

3.10. Density-thinning controls

To test whether the observed β values reflect genuine exclusion structure rather than point density, the 2D prime embedding ($N = 10^5$, $\rho = 0.096$, $\beta = 2.15$) was randomly thinned to match point counts corresponding to $\rho \in \{0.01, 0.02, 0.032, 0.05, 0.096\}$ (20 trials each). At each density, β was compared against a density-matched CSR baseline (20 realisations each).

At the PSI density ($\rho = 0.032$), prime β decreased to 1.46 ± 0.02 [1.42, 1.49], and at the CSR calibration density ($\rho = 0.02$) to 1.30 ± 0.05 . Critically, prime β remained significantly above the density-matched CSR baseline at all densities (CSR: $\beta = 0.98 \pm 0.05$ at $\rho = 0.032$; 0.97 ± 0.04 at $\rho = 0.02$; 0.99 ± 0.02 at $\rho = 0.096$), confirming that the prime embedding possesses genuine exclusion structure beyond that expected from density alone. The CSR baseline itself is density-stable across the tested range (0.95–0.99).

Two conclusions follow. First, β captures exclusion strength rather than point density: the CSR baseline is flat, while structured data lie above it at all densities. Second, absolute β values are density-dependent: the same underlying exclusion mechanism (prime gap constraints) produces $\beta = 2.15$ at $\rho = 0.096$ but $\beta = 1.46$ at $\rho = 0.032$. Cross-domain comparisons of absolute β therefore require density matching. The synthetic hard-core calibration (Section 3.4) is density-referenced through the normalisation $r_{\text{excl}} / \langle \text{NN} \rangle$, which provides a density-invariant measure of exclusion strength.

3.11. 2D prime embeddings: embedding geometry determines exclusion

Prime-number embeddings provide a mathematically controlled benchmark: any exclusion signal must originate from arithmetic divisibility constraints rather than physical interactions. Whether the Brody exponent detects such arithmetic exclusion—and whether this depends on how the 1D prime sequence is mapped to 2D—is tested here.

β was computed under three deterministic 2D embeddings at $N = 10^5$: row-major ($\beta = 2.15$), Ulam spiral ($\beta = 1.87$), and Cantor pairing ($\beta = 1.40$). A spatially random Bernoulli control at matched density (30 realisations) yields $\beta_{\text{Bern}} = 1.47 \pm 0.02$.

The Cantor null result. Under Cantor pairing, primes yield $\beta = 1.40$, which is slightly *below* the Bernoulli baseline ($\Delta\beta = -0.07$, two one-sided test for equivalence against Bernoulli: $p_{\text{TOST}} < 0.01$ at equivalence bound ± 0.1). This is a decisive finding: the same prime sequence that produces strong exclusion under row-major embedding ($\beta = 2.15$) produces no detectable exclusion under Cantor pairing. The conclusion is unambiguous: **2D exclusion in prime embeddings is not an intrinsic property of the prime sequence; it is created or destroyed by the embedding geometry.** The row-major embedding amplifies the prime gap structure because consecutive integers—which cannot both be prime (for $n > 2$)—map to horizontally adjacent array positions, translating the 1D gap constraint into 2D nearest-neighbour exclusion. Ulam spiral produces intermediate exclusion ($\beta = 1.87$) by concentrating primes along quadratic-residue diagonals while leaving off-diagonal regions depleted. Cantor pairing destroys this structure by sweeping the array diagonally, decorrelating the prime gap constraint from 2D adjacency. The embedding dependence indicates that β for arithmetic sequences reflects both the 1D sequence statistics and the 2D embedding geometry; claims about “prime point processes” must therefore specify the embedding. Figure 4 summarises these results.

Sparse-integer control: sparsity versus arithmetic structure. To test whether the elevated β for row-major primes ($\beta = 2.15$) reflects the arithmetic constraint of primality or merely the sparsity of the prime set ($\pi(10^5) = 9\,592$, $\rho = 0.096$), a sparse-integer control was performed: $\pi(N)$ random integers uniformly distributed in $[1, N]$ were embedded in the identical row-major geometry and β was computed (30 trials). The sparse-integer control yields $\beta = 1.47 \pm 0.02$ [1.43, 1.50], which is indistinguishable from the Bernoulli CSR baseline ($\beta = 1.46 \pm 0.02$) and substantially below the prime value ($\Delta\beta = +0.68$). This result establishes that the elevated β for row-major primes is a genuine arithmetic signal—not a sparsity artefact—and that the CSR baseline for binary fields at this density ($\rho = 0.096$) is $\beta_{\text{CSR}}^{(\text{binary})} \approx 1.46$, distinct from the continuous 2D CSR baseline of $\beta = 0.96 \pm 0.15$ (Section 3.4). The binary CSR baseline shift reflects the discrete-grid NNS distribution at low fill fraction; all comparisons between binary arithmetic surfaces and their null models use the appropriate binary CSR reference. Figure 5 presents the density-thinning and sparse-integer control results.

3.12. Peak-detection sensitivity

To assess the robustness of β to the peak-detection threshold, a synthetic surface with 50 embedded Gaus-

sian peaks (known positions, amplitudes 0.5–2.0, additive Gaussian noise $\sigma = 0.1$) was analysed under prominence thresholds ranging from 1% to 15% of the height range. The Brody exponent is stable across the 1%–11% range ($\beta = 1.26$ – 1.27 , variation < 0.01), then decreases monotonically to $\beta = 1.07$ at 15% as genuine peaks are increasingly excluded (peak count drops from 425 to 63). The plateau region (1%–11%) corresponds to thresholds that retain all genuine peaks while suppressing noise; the 3% threshold used throughout this work lies within this stable plateau. A systematic sensitivity analysis across all 58 empirical surfaces remains to be performed; the present five-surface analysis (Section 3.1) and synthetic validation establish plausible robustness but do not fully characterise the threshold-dependence of β for arbitrary surface textures. Figure 6 provides the complete sensitivity profile.

3.13. Proof-of-concept: interferometric profilometry

A proof-of-concept application of the Brody pipeline to phase-extracted interferometric profilometry of a certified roundness standard (JENOPTIK FN 111) is presented in Supplementary Information (SI, Section S4). The concatenated five-revolution profile (629 peaks) yields $\beta = 2.00$ [1.90, 2.10]; per-revolution instability (5–9 peaks, $\beta = 1.00$ – 3.68) qualifies the result as exploratory. Scale-invariance analysis of $\Delta D_2(N)$ for primes across $N \in [10^4, 10^7]$ is also reported in SI (Sections S1.5–S1.6).

4. Discussion

4.1. What β measures—and what it does not

The β – r_{excl} mapping is empirical. The relationship between the Brody exponent and the normalised exclusion radius is established through calibration against synthetic hard-core and Strauss soft-core point processes (Section 3.2). It is not derived from first principles. The mapping is monotonic and reproducible, which makes it practically useful, but it should be understood as an empirical calibration curve rather than a theoretical law. Whether a deeper connection exists between the Brody functional form and 2D exclusion physics—for instance, through determinantal point-process kernels [29]—is an open question.

The calibration provides a physical interpretation: $\beta \approx 0.96$ corresponds to no exclusion (CSR), $\beta \approx 2$ to Ginibre-level repulsion ($r_{\text{excl}} \approx 0.55 \langle \text{NN} \rangle$), and $\beta \gtrsim 5$ to near-crystalline exclusion ($r_{\text{excl}} \gtrsim 0.8 \langle \text{NN} \rangle$). Values of $\beta > 2$ are not interpreted as higher-order RMT symmetry classes; they serve solely as a phenomenological

measure of exclusion strength. Conceptually, the β - r_{excl} relation plays a role analogous to the hard-sphere equation of state [30]: both compress complex spatial correlations into a single interpretable quantity. Unlike the Carnahan–Starling equation, however, the β - r_{excl} calibration is purely empirical.

The effective hard-core radius r_{eff} —the 5th percentile of the NN distance distribution—provides independent validation. Across the synthetic calibration data, $r_{\text{eff}}/\langle \text{NN} \rangle$ correlates with β at Spearman $\rho = 0.988$ ($p < 10^{-4}$). This near-perfect correlation establishes that β and the effective exclusion radius capture essentially identical information, providing the quantitative link between Brody-NNS analysis and standard spatial point-process theory [14].

Critically, β does *not* identify the generative process, replace $g_2(r)$, detect long-range order, or serve as a general-purpose surface descriptor. Adding β to ISO 25178 parameters did not improve manufacturing-process classification accuracy (Section 3.1). This null result is expected: β measures short-range exclusion, while manufacturing processes differ in many aspects (roughness amplitude, skewness, waviness, material).

4.2. The two CSR baselines

A necessary finding is that the 2D CSR baseline is $\beta = 0.96 \pm 0.15$, not $\beta = 0$. This correction is essential: the 2D NNS distribution under CSR is Rayleigh, $P(s) = (\pi s/2) \exp(-\pi s^2/4)$, which is not a member of the Brody family. The Brody distribution at $\beta \approx 0.96$ provides an excellent approximation (KS distance ≈ 0.05), outperforming the analytic Rayleigh in 22 of 30 CSR trials. The baseline is density-stable ($\beta_{\text{CSR}} = 0.95$ – 0.99 across $\rho \in [0.01, 0.10]$).

A second, distinct CSR baseline emerges for binary fields at low fill fraction. For the Bernoulli null at $\rho = 0.096$ (matching the prime surface density), $\beta_{\text{CSR}}^{(\text{binary})} = 1.46 \pm 0.02$, substantially above the continuous CSR value. This shift reflects the discrete-grid NNS distribution at low occupancy. The practical implication is summarised in Table 2.

Table 2: CSR baseline selection for 2D Brody analysis.

Data type	CSR baseline	Example
Continuous 2D	$\beta_{\text{CSR}} = 0.96 \pm 0.15$	FV surfaces, PSI
Binary, $\rho < 0.2$	Density-matched Bernoulli	Primes, arithmetic
Uncertain	Generate own (20 reps)	Any new class

Using the wrong baseline produces spurious exclusion claims. All comparisons in the present work use density-matched CSR references.

4.3. Density dependence and cross-domain comparison

Density-thinning experiments (Section 3.6) reveal that β captures exclusion strength rather than point density—the CSR baseline is flat, while structured data consistently exceed it—but that absolute β values are density-dependent. The 2D prime embedding yields $\beta = 2.15$ at native density $\rho = 0.096$ but $\beta = 1.46 \pm 0.02$ when thinned to the PSI density ($\rho = 0.032$). This density dependence has two implications.

First, it resolves the apparent convergence of primes and PSI at $\beta \approx 2.0$: at matched density, primes ($\beta = 1.46$) and PSI ($\beta = 2.00$) occupy distinct regions of the β scale. The similarity of their native-density β values is partially a density artefact, not evidence of identical exclusion mechanisms. Both domains nevertheless exhibit β values significantly exceeding their respective CSR baselines, confirming genuine exclusion structure.

Second, and more constructively, the synthetic hard-core calibration (Section 3.4) uses the normalised exclusion radius $r_{\text{excl}}/\langle \text{NN} \rangle$, which is density-invariant by construction. Cross-domain comparisons should therefore be performed in the $(r_{\text{excl}}/\langle \text{NN} \rangle, \beta)$ parameter space rather than by comparing absolute β values. When referenced to this density-invariant calibration, all three domains occupy a consistent region of the parameter space: exclusion radii in the range 0.3–0.7 (NN), corresponding to $\beta \approx 1.3$ – 4.6 depending on density.

4.4. Physical interpretation by manufacturing process

In the present dataset, burnished surfaces exhibited the highest median β ($\beta = 1.3$ – 4.3 , $r_{\text{excl}}/\langle \text{NN} \rangle \approx 0.5$ – 0.75). Surfaces produced by turning and grinding ($\beta = 0.0$ – 2.2 , $r_{\text{excl}}/\langle \text{NN} \rangle < 0.55$) spanned from CSR-indistinguishable to moderate exclusion. Milled surfaces showed the widest β range ($\beta = 0.0$ – 3.7). Whether these differences reflect the manufacturing process, the material, or their interaction cannot be determined from the present observational design, in which process and material are confounded (e.g., all burnished surfaces are ductile metals). A factorial experiment crossing process $\times$ material is required for causal attribution. The qualitative observation that finishing operations produced higher β than abrasive operations in the present dataset was robust to peak-detection algorithm choice and threshold variation.

4.5. Internal consistency with $g_2(r)$

The pair correlation function $g_2(r)$ provides an internal consistency check: the exclusion radius inferred from β (via the hard-core calibration) and from the $g_2(r)$ contact value (smallest r where $g_2(r) > 0.5$) are systematically related across 7 FV surfaces (ratio 0.24 ± 0.17). The systematic offset reflects the different definitions of “exclusion”: β captures the shape of the full NNS distribution, while the $g_2(r)$ threshold is a single-point measure at contact. Both measures consistently identify the same surfaces as having high or low exclusion, supporting the interpretation of β as an exclusion measure.

4.6. Complementary descriptors

Binary patterns where peak detection is inapplicable are characterised via D_2 correlation dimension and Banach-space invariants (SI, Sections S1.1–S1.7). These descriptors are empirically orthogonal to β (Pearson $r = -0.31$), confirming that β probes a specific axis—exclusion strength—within a broader spatial-order parameter space. Square-free numbers and the Liouville function serve as negative controls indistinguishable from the Bernoulli null, while primes exhibit $\Delta D_2 > 0$ driven by the fundamental theorem of arithmetic (consecutive integers cannot both be prime).

A noteworthy tension exists between the D_2 and β results. D_2 for primes is stable across row-major, Ulam, and Cantor embeddings ($\Delta D_2 \approx +0.05$ – 0.09 ; SI, Section S1.1), indicating that the multi-scale mass distribution is an intrinsic property of the prime sequence. In contrast, β varies from 2.15 (row-major) to 1.40 (Cantor), indicating that short-range exclusion is embedding-dependent. This tension is resolved by recognising that D_2 and β probe fundamentally different spatial scales: D_2 responds to mass distribution across all box sizes simultaneously, whereas β is dominated by the shortest nearest-neighbour distances. An embedding that preserves the prime gap constraint at nearest-neighbour scales (row-major) yields high β ; an embedding that decorrelates the gap constraint from 2D adjacency (Cantor) yields β indistinguishable from CSR, even though the multi-scale arithmetic structure remains detectable by D_2 . The complementarity of β , D_2 , and Banach descriptors suggests that spatial point patterns may be classified by their coordinates in a low-dimensional descriptor space whose axes correspond to physically interpretable order parameters.

4.7. Limitations

Five limitations qualify the present findings. (1) The β - r_{excl} calibration is empirical, not derived from first

principles; the Brody functional form is justified by AIC comparison (Section 3.2) but its optimality for 2D exclusion is not established analytically. (2) Absolute β values are detector-dependent: the watershed detector recovered 37 grid-limited surfaces but shifted absolute β values (range 0.71–4.61, mean 2.89 ± 1.10); relative ordering is robust. A systematic threshold sensitivity analysis across all 58 surfaces is needed. (3) Surfaces with $n_{\text{NNS}} < 30$ have wide confidence intervals (± 1.0 – 1.5) and are indicative only. (4) The PSI result is exploratory: a single standard, 5–9 peaks per revolution, per-revolution β spanning 1.00–3.68. Replication on additional standards is required. (5) The surface dataset is observational (process $\times$ material confounded); causal attribution requires a factorial experiment. Test–retest reliability has not been established. (6) The β scale is calibrated only on the anti-clustered side (CSR through near-crystalline). The behaviour of β for clustered point processes (Thomas, Matérn) is not established here; β values significantly below the CSR baseline may indicate clustering, but this requires dedicated study with appropriate benchmarks.

4.8. Outlook

The calibrated framework enables several directions. In spatial statistics, connecting β to formal point-process model parameters (Strauss γ , area-interaction radius, DPP kernels) would bridge RMT-inspired NNS analysis with established spatial point-process theory [29]. In surface engineering, β may serve as a target parameter where peak proximity is the quantity of interest, subject to test–retest validation. In number theory, the embedding-dependence of β for primes—particularly the Cantor null result—provides an empirical constraint on how 1D arithmetic structure manifests in 2D spatial statistics. Extension to 3D (additive-manufacturing porosity, granular packings) and to additional point-process domains (galaxy distributions, vortex matter, neural spike patterns) would test the generality of the calibration framework.

5. Conclusion

The Brody exponent β , calibrated against the 2D complete-spatial-randomness baseline ($\beta = 0.96 \pm 0.15$, not $\beta = 0$), provides an empirically robust measure of short-range exclusion strength in 2D spatial point processes. Synthetic hard-core and soft-core experiments establish a monotonic β - $r_{\text{excl}}/\langle \text{NN} \rangle$ mapping spanning $\beta \approx 1$ (CSR) to $\beta \approx 10$ (near-crystalline exclusion). This calibration—empirical rather than derived from

first principles—translates the dimensionless Brody parameter into a physically interpretable exclusion radius and provides the correct 2D reference frame for any application of Brody statistics to spatial data. Values of $\beta > 2$ are not interpreted as higher-order RMT symmetry classes but solely as a phenomenological measure of exclusion strength.

Three unrelated generative mechanisms were analysed within this calibrated framework. Manufactured surface peaks ($\beta = 0$ to $+4.6$, 58 surfaces, 10 processes) exhibit process-dependent exclusion; in the present dataset, burnished surfaces showed the highest median exclusion and turned surfaces the weakest ($\beta \approx 0.4$, indistinguishable from CSR). 2D binary embeddings of the primes yield $\beta = 2.15$ (row-major), 1.87 (Ulam), and 1.40 (Cantor) at $N = 10^5$, demonstrating that β depends on the embedding geometry as well as on the arithmetic sequence. At matched density ($\rho = 0.032$), the row-major prime embedding yields $\beta = 1.46 \pm 0.02$, significantly exceeding the density-matched CSR baseline (0.98 ± 0.05) but below the native-density value. Phase-extracted interferometric profilometry of a certified roundness standard—included as an independently generated spatial point process from optical phase reconstruction—yields $\beta = 2.00$ [1.90, 2.10] at $\rho = 0.032$, also well above CSR.

Density-thinning and embedding-robustness experiments establish three findings. (i) β captures exclusion rather than density: the CSR baseline is density-stable (0.95–0.99 across $\rho \in [0.01, 0.10]$) while structured data consistently exceed it. (ii) Absolute β values are density-dependent: cross-domain comparisons require density matching or use of the density-invariant $r_{\text{excl}}/\langle \text{NN} \rangle$ calibration. (iii) β for 2D arithmetic embeddings is embedding-dependent: the same prime sequence produces different β under different 2D mappings, so claims about the spatial statistics of primes must specify the embedding.

For binary patterns where peak detection is inapplicable, complementary D_2 and Banach-space descriptors capture orthogonal aspects of spatial structure (Pearson $r = -0.31$), with square-free numbers and the Liouville function serving as decisive negative controls (Supplementary Information).

The Cantor embedding null result is particularly instructive: the same prime sequence yields $\beta = 2.15$ (row-major), strong exclusion, and $\beta = 1.40$ (Cantor), no exclusion ($\Delta\beta = -0.07$ relative to the Bernoulli baseline, $p_{\text{TOST}} < 0.01$). This demonstrates that 2D exclusion in arithmetic embeddings is not an intrinsic property of the arithmetic sequence—it is created or destroyed by the embedding geometry. The framework

therefore detects both the presence and the absence of exclusion, which is essential for any measurement tool.

The β – r_{excl} calibration (Spearman $\rho = 0.988$ with r_{eff}), the 2D CSR baseline correction ($\beta = 0.96 \pm 0.15$ for continuous processes; $\beta_{\text{CSR}}^{(\text{binary})} \approx 1.46$ for binary fields at low density), and the density-thinning and embedding-robustness protocols together constitute a calibrated measurement framework for systematic, reproducible characterisation of short-range exclusion in 2D spatial point processes. The framework is applicable across physical, mathematical, and engineering domains, provided the necessary corrections—density matching, embedding specification, and reference to the appropriate CSR baseline—are applied.

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Supplementary Information

S1. Extended validation of D_2 and Banach descriptors

The main text (Section 3.6–3.7) presents the essential findings from the D_2 and Banach-space descriptor analysis. This Supplementary Information provides the complete set of validation results.

S1.1 Embedding independence. The D_2 signal for binary primes persists across three deterministic 2D embeddings (row-major, Ulam spiral, Cantor diagonal) with $\Delta D_2 = +0.053$ to $+0.089$, ruling out embedding-geometry artefacts. Under random permutation of the 1D-to-2D mapping (30 surrogates), D_2 drops from 1.687 to 1.642 ± 0.003 ($z = 13.5$), confirming the signal is intrinsic to the prime sequence.

S1.2 Estimator bias. The box-counting estimator of D_2 exhibits finite-size bias: $2 - D_2^{(\text{Bern})} \approx 12.9 / \log N - 0.86$. At $N = 10^5$, this accounts for $\sim 250\%$ of the apparent deviation of $D_2^{(\text{prime})}$ from 2. The bias-cancelled excess $\Delta D_2 \approx 0.04$ – 0.05 is stable across $N \in [10^4, 10^7]$.

S1.3 Controlled-correlation ordering. A calibrated scale of surrogate binary sequences at matched prime density establishes: $D_2(\text{block-clustered}) = 1.504 < D_2(\text{Bernoulli}) = 1.645 < D_2(\text{primes}) = 1.687 < D_2(\text{repulsive}) = 1.797$, confirming that primes occupy an intermediate anti-clustered position.

S1.4 Arithmetic sequences beyond primes. The D_2 analysis was applied to additional arithmetic sequences at $N = 5 \times 10^4$ – 10^5 , selected to span a range of multiplicative constraints: twin primes (upper), primes $\equiv 1 \pmod{4}$ and $\equiv 3 \pmod{4}$ (Chebyshev bias test), sums of two squares, and square-free numbers (negative control). The Möbius $\mu^\pm(n)$ and Liouville $\lambda(n)$ functions were also tested. All sequences except square-free numbers and the Liouville function survive Benjamini–Hochberg correction (FDR = 0.05). Square-free numbers and the Liouville function are indistinguishable from the Bernoulli null, serving as negative controls. Beatty sequences $[an] \bmod 2$ for selected irrationals are discussed in S1.5; binary digit sequences of fundamental constants and the Champernowne and Copeland–Erdős constructions did not produce statistically significant ΔD_2 after bias correction and are omitted for brevity.

S1.5 Convergence analysis. $\Delta D_2(N)$ for primes persists across $N \in [5 \times 10^3, 10^6]$, while Möbius μ^- fluctuates around zero and Beatty φ oscillates with Moiré alignment resonances.

S1.6 Analytic bias function. An analytic approximation for the box-counting bias in sparse binary surfaces is derived: $D_2(N, p) = \log(1/p + W^2/4)/\log(W/4) - \log(1/p + 4)/\log(W/4)$, with empirical correction factor 1.24 ± 0.15 .

S3. Complete Brody β table (all 58 surfaces)

Table S1: Brody exponent β for all 58 analysable surfaces (21 prominence-based + 37 watershed-recovered), ordered by decreasing β .

S2. Supplementary figures

S4. $\beta(N)$ convergence data

The Brody exponent β for the row-major binary prime embedding was computed at twelve values of N : $N = 10^3$ ($\beta = 2.20$), $N = 2 \times 10^3$ ($\beta = 2.10$), $N = 5 \times 10^3$ ($\beta = 1.69$), $N = 10^4$ ($\beta = 1.97$), $N = 2 \times 10^4$ ($\beta = 2.19$), $N = 5 \times 10^4$ ($\beta = 1.94$), $N = 10^5$ ($\beta = 2.15$), $N = 2 \times 10^5$ ($\beta = 2.05$), $N = 5 \times 10^5$ ($\beta = 1.78$), $N = 10^6$ ($\beta = 1.75$), $N = 5 \times 10^6$ ($\beta = 1.86$), $N = 10^7$ ($\beta = 1.26$). The trend is non-monotonic; no simple convergence to a fixed asymptotic value is observed at accessible N . The decrease at $N = 10^7$ may reflect the approach to the sparse limit ($\rho = \pi(N)/N \sim 1/\log N \rightarrow 0$ as $N \rightarrow \infty$).

Data Availability

All analysis code, processed data (JSON format), and figure-generation scripts are archived in

Table 3: Complete β values for all 58 manufactured surfaces.

Material	Proc.	n_{peaks}	n_{NNS}	β	95% CI
C45 steel	RT ^w	199	192	4.61	[3.98, 5.48]
ELLOR	RT ^w	231	229	4.54	[4.15, 5.11]
Graphite	BB ^w	201	198	4.49	[4.03, 5.21]
MO58A brass	Bn	41	41	4.30	[3.38, 5.81]
Al7075	WEr ^w	209	208	4.26	[3.80, 4.73]
Ti6Al4V	WEr ^w	228	225	4.16	[3.72, 4.71]
P1-1.4301-t.zgrub	^w	197	178	4.04	[3.64, 4.54]
1.4301 steel	Ho	49	49	4.00	[3.22, 5.12]
ELLOR	Tu ^w	198	195	3.99	[3.52, 4.71]
Al7075	RT ^w	135	116	3.97	[3.54, 4.39]
Ti6Al4V	WEf ^w	181	179	3.91	[3.44, 4.66]
ELLOR	RM ^w	196	195	3.90	[3.51, 4.39]
MO58A brass	WEr ^w	186	180	3.89	[3.52, 4.36]
Al7075	Mi	44	43	3.70	[2.94, 4.91]
Ti6Al4V	Tu ^w	107	99	3.68	[3.28, 4.18]
C45 steel	RM ^w	111	109	3.58	[3.09, 4.35]
C45 steel	Tu ^w	139	132	3.34	[2.99, 3.78]
C45 steel	WEr ^w	180	180	3.30	[2.88, 3.86]
1.4301 steel	WEr ^w	147	145	3.28	[2.93, 3.83]
Ti6Al4V	Mi ^w	133	130	3.25	[2.78, 3.84]
C45 steel	WEf ^w	178	176	3.20	[2.83, 3.74]
1.4301 steel	Mi	29	18	3.10	[2.03, 4.86] [†]
Al7075	Bn	30	29	3.10	[2.35, 4.49] [†]
Graphite	Gr ^w	132	131	3.03	[2.65, 3.71]
Al7075	WEf ^w	97	92	2.98	[2.48, 3.62]
ELLOR	WEr ^w	88	87	2.78	[2.28, 3.70]
MO58A brass	RM ^w	71	68	2.70	[2.36, 3.24]
Ti6Al4V	Bn	28	28	2.60	[1.85, 3.86] [†]
1.4301 steel	WEf ^w	98	95	2.59	[2.20, 3.14]
MO58A brass	Tu ^w	30	25	2.46	[1.80, 3.70] [†]
Al	BB	25	23	2.40	[1.68, 3.74] [†]
Al	Ho ^w	110	110	2.26	[1.88, 2.69]
1.4301 steel	RM ^w	100	99	2.25	[2.01, 2.59]
1.4301 steel	Gr	32	21	2.20	[1.44, 3.46] [†]
Ti6Al4V	RT ^w	120	120	2.20	[1.94, 2.59]
C45 steel	Ho ^w	37	35	2.18	[1.53, 3.36]
Brass	Ho ^w	72	71	2.00	[1.64, 2.50]
1.4301 steel	Bn	68	65	1.80	[1.36, 2.38]
C45 steel	Gr ^w	24	21	1.73	[1.15, 2.74] [†]
Brass	BB ^w	82	82	1.71	[1.36, 2.32]
C45 steel	Mi ^w	25	22	1.64	[1.24, 2.43] [†]
Ti6Al4V	Ho	32	27	1.40	[0.87, 2.34] [†]
C45 steel	BB ^w	14	14	1.34	[1.09, 1.72] [†]
Al7075	Tu	34	33	1.30	[0.87, 2.02]
C45 steel	Bn	89	88	1.30	[0.99, 1.72]
Al	Gr	169	167	1.20	[0.99, 1.46]
ELLOR	Mi ^w	42	39	1.18	[0.78, 1.84]
ELLOR	WEf ^w	21	17	1.14	[0.79, 1.60] [†]
Graphite	Ho	54	52	0.90	[0.58, 1.32]
Brass	Gr ^w	39	38	0.85	[0.56, 1.36]
MO58A brass	WEf	86	61	0.80	[0.52, 1.21]
1.4301 steel	BB ^w	21	17	0.71	[0.36, 1.22] [†]
MO58A brass	RT	335	279	0.70	[0.56, 0.87]
1.4301 steel	Tu	458	428	0.40	[0.30, 0.52]
16 Ti6Al4V	RM	109	84	0.40	[0.19, 0.66]
MO58A brass	Mi	183	180	0.00	[-0.10, 0.12]
Ti6Al4V	BB	97	94	-0.10	[-0.22, 0.07]
Ti6Al4V	Gr	98	96	-0.20	[-0.31, -0.05]

[†] $n_{\text{NNS}} < 30$: CI width > 1.5, estimate indicative only. ^w Watershed

a public GitHub repository at <https://github.com/dawidkucharski/brody-2d-exclusion>. The repository includes the complete 58-surface Brody- β dataset, all synthetic point-process calibration data, and a reproducible computational environment specified via `requirements.txt`. Surface metrology raw data (focus-variation `.sur` files, PSI interferogram frames) are available from the author upon reasonable request. The `aurora_lab` Python package (v2.0.0) provides reproducible implementations of the surface construction, null-model generation, box-counting, and validation procedures described in this work.

Unified pair correlation and Brody spacing statistics
Manufactured surfaces × prime embeddings × quantum-chaos reference ensembles

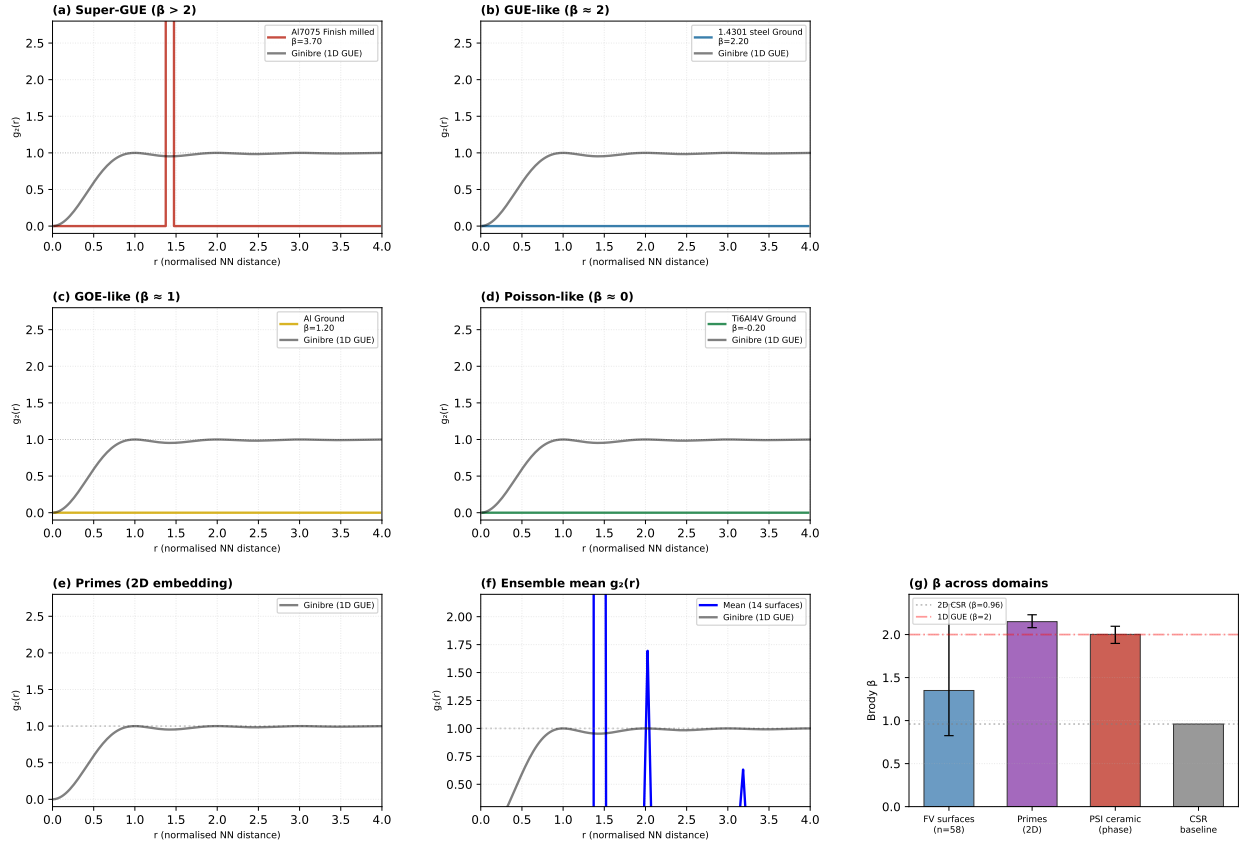


Figure 3: Pair correlation function $g_2(r)$ across domains. (a–d) Representative manufactured surfaces spanning the Brody β range, compared with the Ginibre prediction (solid black). β values are from Table 1. (e) 2D binary embedding of the primes at $N = 10^5$ ($\beta = 2.15$) and matched-density Bernoulli null. (f) Ensemble-mean $g_2(r)$ across 14 manufactured surfaces ($\pm 1\sigma$ band). (g) Brody exponent β across the three domains at native density. Density-thinning controls (Section 3.6) show that absolute β values are density-dependent; cross-domain comparison requires density matching.

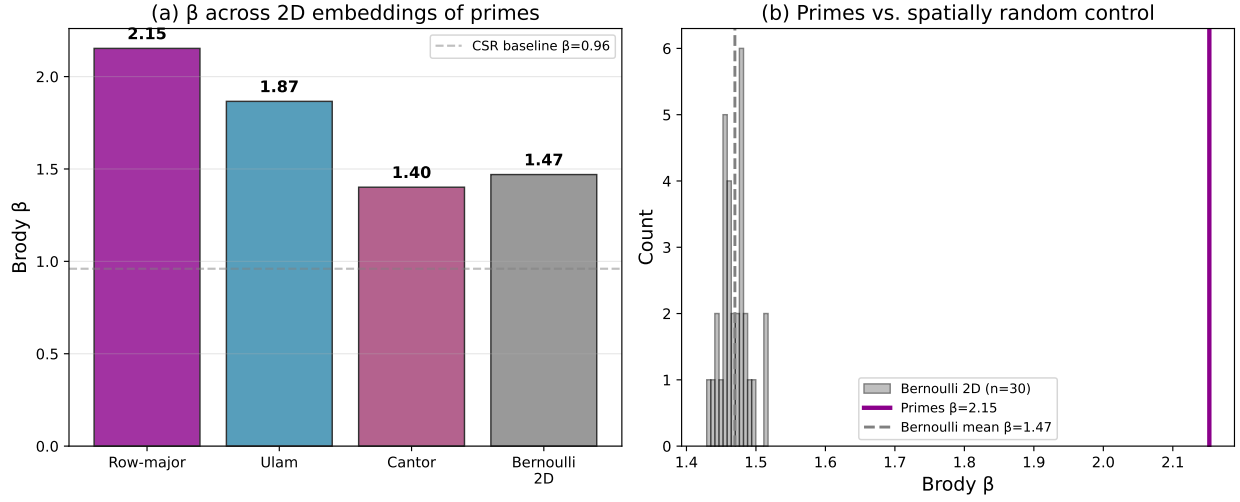


Figure 4: Embedding dependence of the Brody exponent for 2D primes at $N = 10^5$. (a) β under three deterministic embeddings and the Bernoulli 2D control (matched density). Row-major and Ulam embeddings produce β significantly above the spatially random control; Cantor pairing does not. (b) Distribution of β for 30 Bernoulli 2D realisations, with the row-major prime value indicated. The embedding dependence demonstrates that β for arithmetic sequences reflects both sequence structure and embedding geometry.

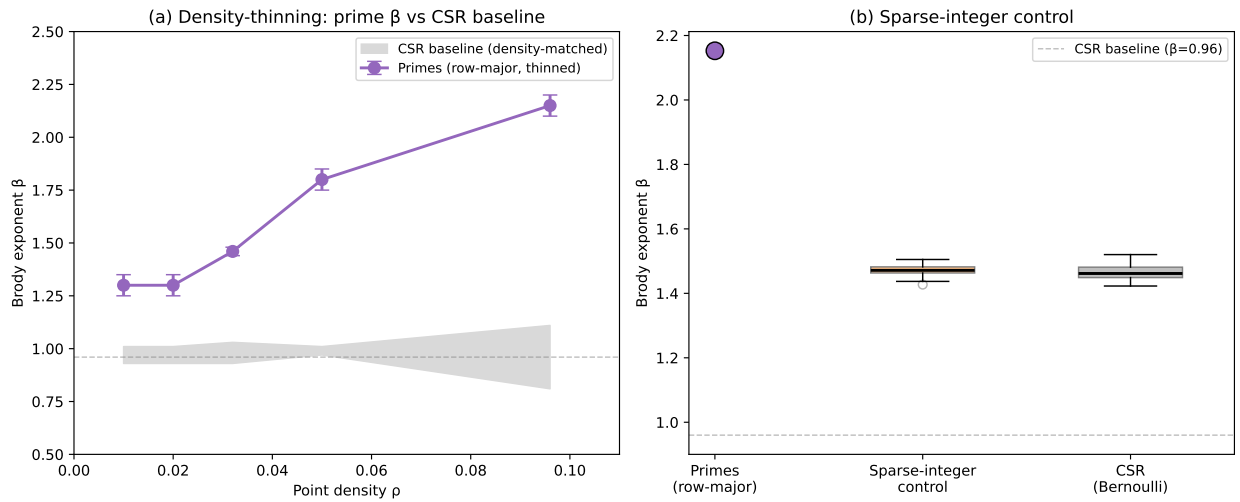


Figure 5: Density-thinning and sparse-integer controls. (a) Brody exponent β for row-major primes thinned to five densities ($\rho = 0.01$ – 0.096), compared with density-matched CSR baselines (grey band). β decreases with density ($\beta = 2.15$ at $\rho = 0.096$ to $\beta = 1.30$ at $\rho = 0.01$) but remains above CSR at all densities. (b) Sparse-integer control: 30 trials of $\pi(N)$ random integers embedded identically to primes yield $\beta = 1.47 \pm 0.02$, indistinguishable from the Bernoulli CSR baseline (1.46 ± 0.02) and substantially below the prime value ($\beta = 2.15$). The prime β signal is therefore a genuine arithmetic effect, not a sparsity artefact.

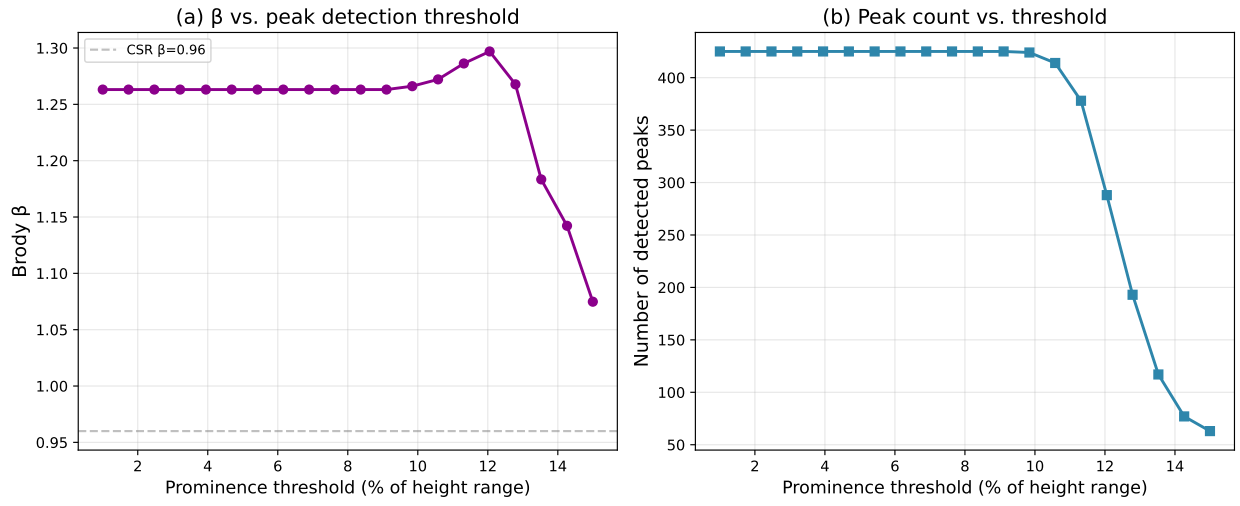


Figure 6: Peak-detection sensitivity analysis on a synthetic surface with 50 embedded Gaussian peaks. (a) Brody β versus prominence threshold (% of height range). β is stable (< 0.01 variation) across 1%–11% thresholds, then decreases as genuine peaks are excluded. (b) Number of detected peaks versus threshold. The 3% threshold used throughout this work (dashed line) lies within the stable plateau.

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